WindBrowser manual

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1 WindBrowser

This manual describes WindBrowser 0.9.63-b.

WindBrowser is an Android app that downloads wind forecasts from the Finnish Meteorological Institute (FMI) for map areas you choose, and **keeps every forecast run** on the phone. It is meant for judging how far a forecast can be trusted, for example before a sail race or a cruise leg on the Baltic Sea.

1.1 Why keep every run?

FMI serves only the latest run of each model. When a new run replaces the old one you cannot see how the forecast has changed. WindBrowser stores each run it downloads, so you can compare successive runs for the same place and time. If they agree, the forecast is probably reliable; if they differ a lot, it is not.

History exists only from the moment the app starts downloading, so add your area a few days before you need it.

1.2 Models

Model	Resolution	Area size limit	Data per square degree and run
HARMONIE	fine (about 2.5 km)	16 square degrees	about 385 kB
ECMWF-IFS	coarse (0.25 degree)	400 square degrees	about 2 kB

HARMONIE covers northern Europe. ECMWF-IFS works anywhere in the world. They can be combined: ECMWF-IFS under HARMONIE gives a wide overview with detail where you need it.

1.3 How to read this manual

The chapters follow the order in which you will use the app:

1. First start: choose your first area.
2. The map: the wind layer and controls.
3. Charts: tap a point and compare runs.
4. Tiles: manage your areas.
5. Stations: observed wind.
6. Settings
7. Updates and troubleshooting
8. About and credits

!!! note The app is for use over the sea. Observation stations are a curated set of marine stations.

2 First start

When you open WindBrowser for the first time the map is empty. **Nothing is downloaded until you have chosen an area.**

The menu (hamburger button, top left) holds **Add area here…** at the top:

2.1 Add an area

1. Open the menu with the **hamburger button** at the top left of the map.
2. Pick **Add area here…**.
3. Choose the **model**: HARMONIE (detailed, small areas) or ECMWF-IFS (coarse, large areas, very small downloads).
4. A rectangle appears inside the current view. Drag its **top-left and bottom-right handles** to cover the sea area you want. Touches elsewhere pan the map.

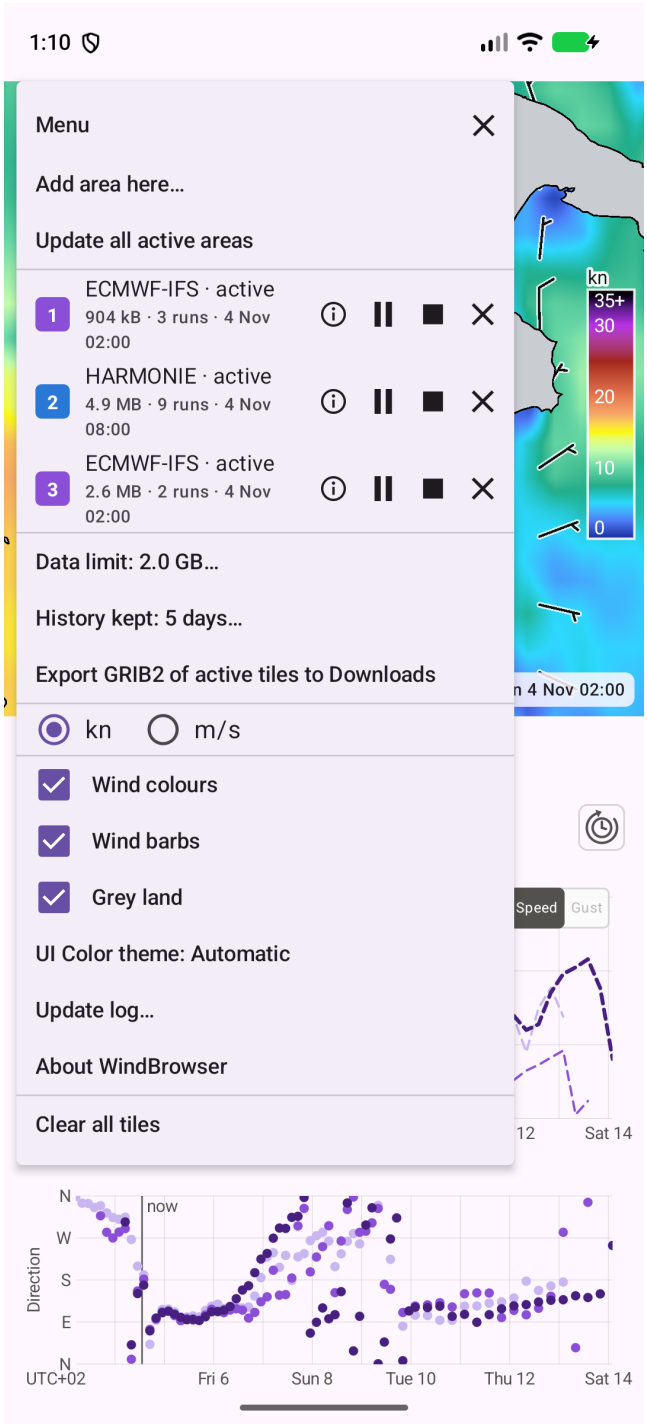


Figure 1: Menu

5. The dashed outline around your rectangle is the area that will really be downloaded. It is slightly larger so that points at the edge of your rectangle can still be interpolated.
6. Check the **disk estimate** in the panel. A warning appears if the data limit would be exceeded.
7. Confirm. The first forecast starts downloading ("Downloading the first forecast…"). When data is ready, the panel under the map says "Tap the map, or an orange station").

!!! tip For a sail race or just a cruise leg, choose a HARMONIE area around the course and a larger ECMWF-IFS area around it. Both can be active at once.

2.2 Size limits

An area can hold at most 16 square degrees with HARMONIE and 400 with ECMWF-IFS. Larger rectangles are refused. See Tiles for what happens when areas overlap.

2.3 What happens next

The app checks for a new forecast every hour in the background, and also when you bring it to the front (at most every 10 minutes). See Updates.

3 The map

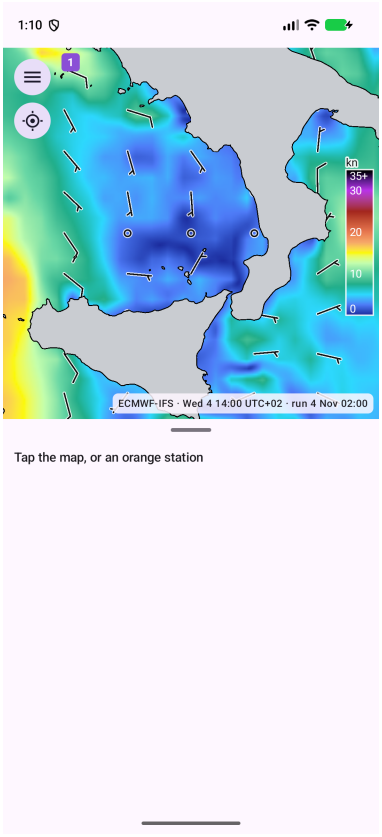
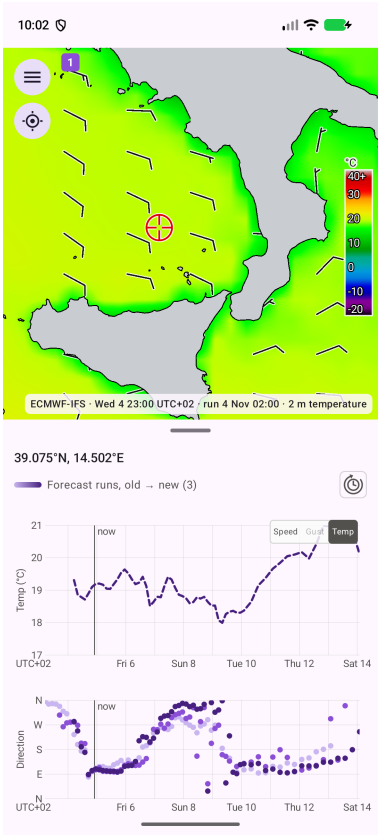


Figure 2: Map overview

3.1 What you see

- **Wind layer**: colours show wind speed, from blue (calm) through green, yellow, orange and red to purple and black (35 knots and more). Wind barbs show direction and speed in knots. The layer is opaque.
- **Colour bar** with the numbers inside it.

- **Temperature map**: choose **Temp** on the toggle in the speed chart and the colours show the 2 m air temperature from -20 °C (black) through purple, blue, cyan, green, yellow and orange to red and light grey at 40 °C and above (the `nipy_spectral` scale), with its own colour bar in °C. The wind barbs stay. Only ECMWF-IFS has temperature: where a HARMONIE tile lies over an ECMWF-IFS tile, the colours come from ECMWF-IFS and the barbs from HARMONIE; a HARMONIE tile with no ECMWF-IFS under it shows barbs only. The grey land setting is kept as it is, so with grey land on, the temperature shows over the sea only. Choose **Speed** to go back to the wind layer.



- **Coastline**: grey land with a black shoreline, drawn from a bundled vector map, so no map tiles are needed and the map works offline. - **Place names**: small names of larger towns; more appear as you zoom in. - **Station dots**: orange dots are observation stations (Stations); small when zoomed out, larger when zoomed in. - **Tile outlines**: the borders of your areas, with a number badge. Solid means active, dashed means paused or stopped. Blue is HARMONIE, purple is ECMWF-IFS.

Zoom runs from level 5 to 15. Drag to pan, pinch to zoom.

3.2 The caption

The caption at the bottom of the map reads, for example, `ECMWF-IFS · Sat 7 20:00 UTC+02 · run 4 Nov 02:00 · tap for now`: the model, the forecast time shown and the model run it comes from. In the temperature map it also says "2 m temperature" and, where the barbs come from another model, "wind barbs FMI-HARMONIE". For a tapped point it names the model that covers the point, and adds "also ECMWF-IFS" when a second model reaches it too. When a point is selected, tap the caption (**tap for now**) to put the chart cursor at the actual time of day; the map shows the nearest forecast hour.

3.3 Old data fades

A tile whose newest run is more than 24 hours old fades to grey, fully grey at 48 hours. Grey means the data is stale, usually because downloads are paused or failing.

3.4 Two models together

ECMWF-IFS is drawn below HARMONIE, so you see the fine model where it exists and the coarse one around it.

3.5 Location button

The button under the menu button finds your position (one fix; GPS if you allowed precise location), selects that point as if you had tapped it, and centres the map there. The position is not stored or sent anywhere.

3.6 Divider

Between map and charts there is a drag handle. Portrait: map above, charts below. Landscape: map on the left, charts on the right. Each orientation remembers its own split.

4 Charts

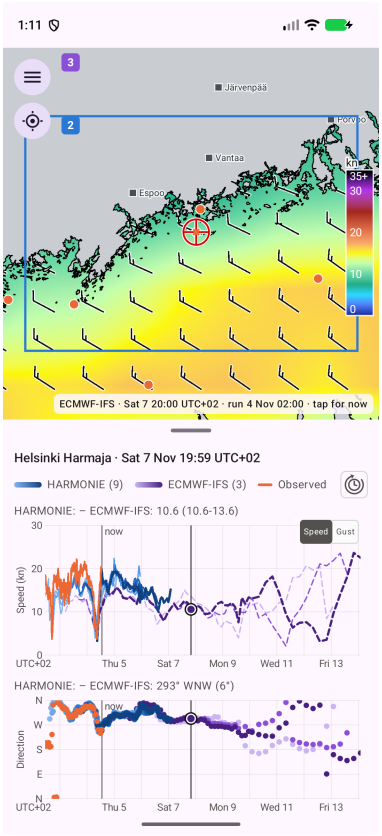


Figure 3: Charts

Tap anywhere on the map to select a point (a red crosshair marks it), or tap an orange station dot; the title then shows the station name or the coordinates. The charts show the forecast at that point from **every stored run** of every area that covers it, whether the tile is active or not.

4.1 Reading the charts

- **Speed chart** (top): each run is one line. HARMONIE is blue and solid; ECMWF-IFS is purple and dashed. Older runs are paler, the newest is darkest. With only one model the legend reads “Forecast runs, old → new”.
- **Direction chart**: shown as dots, never lines, because 359 and 1 degrees are neighbours.

- **Speed / Gust / Temp toggle**: small switch in the corner of the speed chart. Gust exists only for HARMONIE and temperature (2 m air temperature, in °C) only for ECMWF-IFS; each option is greyed out where no run has it. Choosing **Temp** also turns the map's colours to temperature (see The map). If you tap a point that no temperature forecast covers, the chart and the map go back to speed.
- **Observed wind** from a tapped station is drawn in orange. With **Temp** chosen it is the observed air temperature (FMI, SMHI, DMI and Marine Helsinki stations).

If the runs lie close together the forecast is consistent. A wide fan means uncertainty.

4.2 The cursor and the readout

Touch a chart with **one finger** to move the cursor. The map follows the cursor in time. The line above each chart reads, for example:

`HARMONIE: 12.2 (11.1–13.0)  ECMWF-IFS: 15.2 (11.1–15.2)  Observed: 13.2`

Before you touch the chart the models show a dash (–).

For each model it shows the value of the newest run covering that time, and in brackets the lowest and highest of all runs covering it, then the observation.

4.3 Legend

The legend shows each model with its number of runs. **Tap an entry** to hide that model's lines, dots and readout; tap again to show it.

4.4 Zoom

Two fingers zoom and pan the shared time axis. Speed rescales to what is in view and hourly ticks appear when zoomed in. **Tap for now** resets the axis.

4.5 Auto-advance

The clock button with the circular arrow at the right end of the legend moves the cursor, and so the map, forward by itself: a quarter of an hour per quarter second, from the cursor (or from now) to the end of the axis in view. A touch on a chart, the button again, or "now" stops it.

4.6 Units

Speed is shown in knots or m/s (Settings). Times show the phone's UTC offset, for example `UTC+02`.

5 Tiles

An area you download is called a **tile**. You can have up to **nine**, numbered 1 to 9. Everything is managed from the hamburger menu.

5.1 The tile list

Each row shows the tile's number (blue badge HARMONIE, purple ECMWF-IFS), model, state, size, number of runs and newest run, for example `HARMONIE · active · 4.9 MB · 9 runs · 4 Nov 08:00`. Tap a row to zoom the map to that tile.

Button	Action
i	List the stored runs (model run time and file size)
pause / play	Pause stops downloads but keeps the data; play resumes

Button	Action
stop	Stops downloads; the tile is removed when its data has expired
delete	Removes the tile and its data. Always asks for confirmation

States: **Active** downloads; **Paused** keeps data but does not download; **Stopped** is removed once its data has expired.

5.2 Overlap

Only active tiles of the same model count.

- An area completely covered by existing tiles is refused.
- If 70 % or more is covered by one tile, the app offers to **enlarge that tile** to hold both, or to add the area anyway.
- A smaller overlap is accepted with a note that the overlap is downloaded twice.

5.3 Storage limits

- **Data limit**: 1 to 5 GB (default 2). When exceeded, the oldest runs go first, those of non-active tiles before active ones; the newest run of an active tile is never removed. A dialog tells you once what was removed and what you can do: shorter history, smaller areas or fewer tiles, ECMWF-IFS, or a higher limit.
- **History**: 1 to 14 days.

Clear all tiles always asks for confirmation.

6 Stations

The small orange dots on the map are marine observation stations. Tap one to see its **measured wind** (and, where the station has a thermometer, its **air temperature** on the Temp chart) drawn on the charts together with the forecasts, so you can see how well recent runs matched reality.

6.1 Sources

Source	Data
FMI	10-minute observations
Marine Helsinki	Kruunuvuorenselkä
SMHI (Sweden)	hourly
DMI (Denmark)	10-minute; gust is the highest 3-second mean
TalTech	Tallinnamadal, latest values only
Estonian Environment Agency	current wind only, as a text label

Estonian stations and Tallinnamadal show only the wind of the moment as a label on the map, not a chart, because those sources offer no history. Where a source has no gust or direction, those lines are left out. A gap in the line means the station was silent.

Station dots are 2 dp in a wide view and grow to 12 dp from zoom level 10.

7 Settings

All settings are in the hamburger menu.

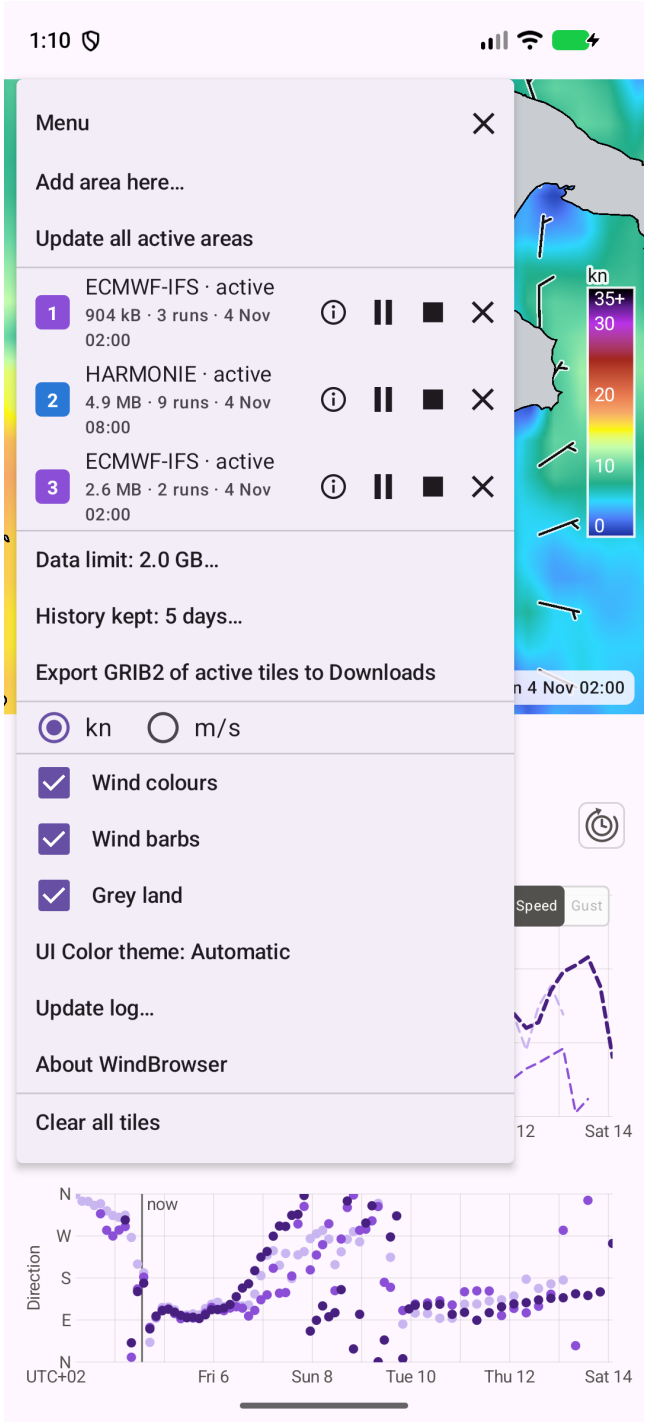


Figure 4: Tile list

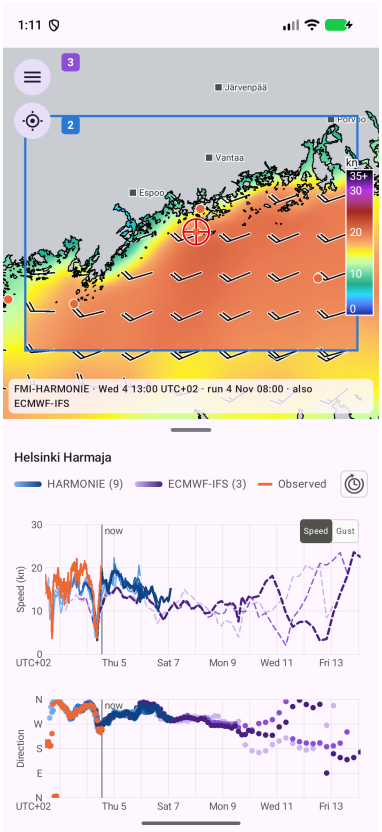


Figure 5: Station

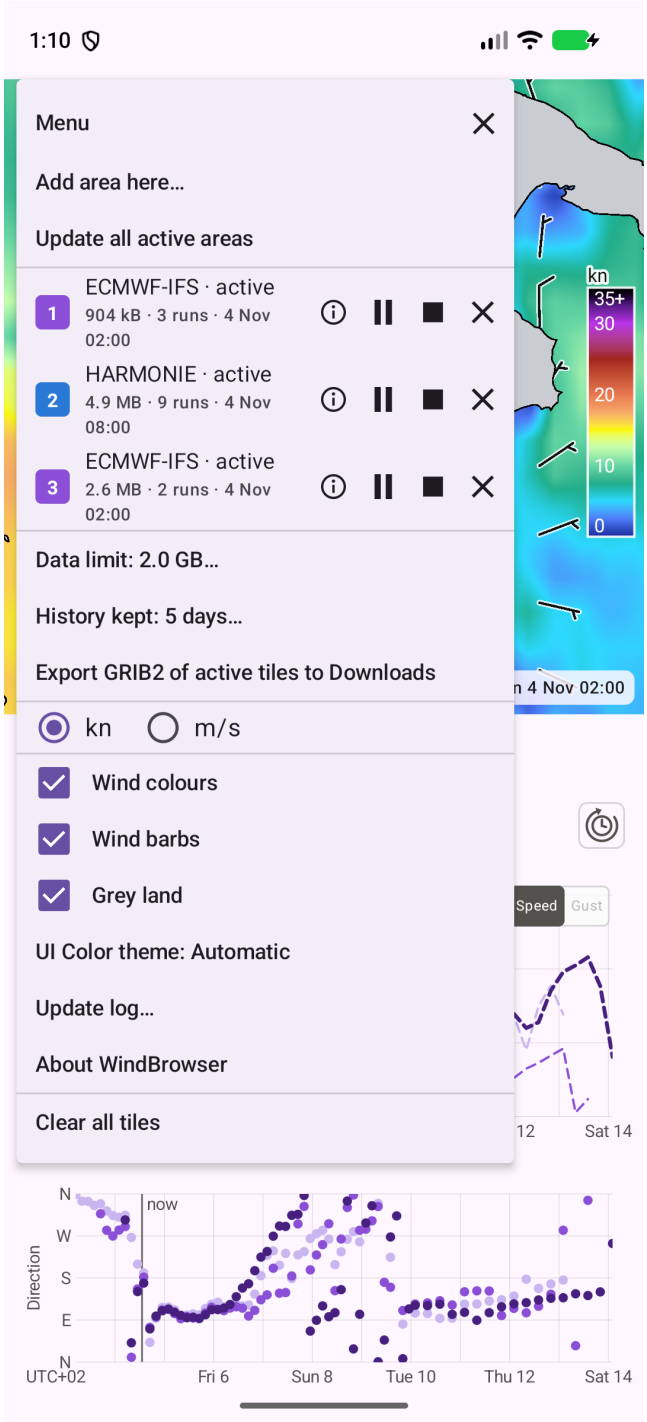


Figure 6: Menu

Setting	Meaning
Update all active areas	Starts an update now
Data limit	1–5 GB (default 2)
History kept	1–14 days of runs to keep (applied immediately)
Export GRIB2 of active tiles to Downloads	Copies the newest raw GRIB2 of every active tile to Downloads, with the tile number in the name (Android 10 and newer)
kn / m/s	Speed unit, default knots. Barbs are always in knots
Wind colours, **Wind barbs**	Tick each layer on or off
Grey land	Untick to show only the shoreline
UI Color theme	Automatic, light or dark; tap to cycle
Update log…	See Updates
Manual	This manual, stored in the app, so it works without a connection
About WindBrowser	Credits, version, and links to the online manual and the PDF
Clear all tiles	Removes everything; always asks first

8 Updates and troubleshooting

8.1 How updates work

- A background job runs about **hourly** and updates every active tile.
- The app also updates when you bring it to the front, at most every 10 minutes.
- FMI keeps **only its latest run**. A run missed because the phone was idle cannot be fetched later, so open the app now and then in the days before you need it.

8.2 Update log

Menu → **Update log…** lists each job (hourly or manual, attempt, standby bucket), the result for each tile, and the end. Use it to find out why a run is missing.

8.3 Troubleshooting

Problem	What to try
"Downloading the first forecast…" does not finish	Check the connection; see the update log
Tile is grey	Its newest run is older than 24 h. Check that it is active and that the phone is online
"No forecast at the tapped point"	The point is outside every tile. Add an area (ECMWF-IFS works anywhere)
Runs disappeared	The data limit or history limit pruned them; raise it
HARMONIE area refused	Larger than 16 square degrees, or fully covered by existing tiles
Android delays hourly updates	Android power saving; open the app to trigger an update

9 About and credits

WindBrowser keeps FMI forecast runs for comparison. The version is shown in the menu under **About WindBrowser**.

9.1 Where to find the manual

- **In the app, offline:** menu → **Manual**.
- **Online (latest):** https://pclusius.github.io/windbrowser-docs/
- **PDF:** https://pclusius.github.io/windbrowser-docs/WindBrowser-manual.pdf, also linked from **About WindBrowser**.

9.2 Data and credits

- Forecasts: **Finnish Meteorological Institute** open data (HARMONIE and ECMWF IFS).
- Observations: FMI, SMHI, DMI, Marine Helsinki, TalTech and the Estonian Environment Agency (ilmateenistus.ee). Licences are listed in the app.
- Coastline: OpenStreetMap contributors (land polygons) and Natural Earth.
- Place names: GeoNames (CC BY 4.0).

No account is needed. Your position, if you use the location button, is never stored or sent.